Induced Matchings in Prescribed Edge Sets of Graphs of Maximum Degree Four

Route A: the paper-first proof, with a matching Lean formalization

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Abstract. Let G be a finite simple graph of maximum degree at most four. We prove that every prescribed set $F \subseteq E(G)$ contains an induced matching M of the original graph such that $|F| \le 20|M|$. Equivalently, every induced subgraph of the square of the line graph of G has an independent set containing at least one twentieth of its vertices. The proof establishes five clique bounds, according to the degrees of their vertices in the full conflict graph, and combines them with a private-neighbourhood counting argument. All structural bounds needed for the argument are proved below. The constant is sharp by the familiar doubled five-cycle. The full-edge-set bound and this extremal example are already present in Joos and Nguyen (2016); the prescribed-set formulation retains all conflicts of the original graph. An appendix identifies the corresponding declarations in the PaperRoute formalization. No claim of priority over the entire literature is made.

Keywords. Induced matching; prescribed edge set; square of a line graph; Hall ratio; maximum degree four; formal verification.

1. Statement and context

All graphs in this manuscript are finite, simple and undirected. An **induced matching** in G is a set of pairwise vertex-disjoint edges such that no edge of G joins endpoints of two different chosen edges. In particular, an edge outside a prescribed candidate set is still allowed to prohibit a pair of candidate edges.

Theorem 1.1 (Prescribed-edge packing). Let $\Delta(G) \le 4$ and let $F \subseteq E(G)$. There is an induced matching M in G satisfying

$$M \subseteq F, \qquad |F| \le 20|M|. \tag{1.1}$$

Thus $|M| \ge \lceil |F|/20 \rceil$. The constant 20 cannot be replaced by any smaller real constant.

For distinct edges $e, f \in E(G)$, say that e and f **conflict** if they share an endpoint or if an endpoint of e is adjacent in G to an endpoint of f. The **conflict graph** H_G has vertex set $E(G)$ and this adjacency relation; it is the square $L(G)^2$ of the line graph. Independent sets of H_G are exactly induced matchings of G. Write $c(e) = d_{H_G}(e)$ for the full conflict degree of an edge.

If $\alpha(J)$ is the independence number of a graph J, its Hall ratio is

$$\rho(J) = \max_{\varnothing \ne X \subseteq V(J)} \frac{|X|}{\alpha(J[X])}. \tag{1.2}$$

We set $\rho(J) = 0$ when J is empty. Theorem 1.1 is equivalent to $\rho(H_G) \le 20$. Here $H_G[F]$ is obtained by restricting the vertices of the conflict graph, while its edges retain their meaning in the full original graph.

Joos and Nguyen [1, Theorem 1.1 and the corollary on p. 156] prove

$$\nu_s(G) \ge \frac{n(G) - i(G) - c_5(G)}{9}, \tag{1.3}$$

where $i(G)$ counts isolated vertices and $c_5(G)$ counts components isomorphic to the doubled five-cycle. They explicitly deduce $\nu_s(G) \ge |E(G)|/20$, and obtain $|E(G)|/18$ when there is no such component. Their

sharpness example and the optimality of the full-edge-set constant are therefore prior results. Theorem 1.1 extends the full-edge-set 20-bound to prescribed sets; it does not subsume all the more refined assertions in (1.3).

The distinction between these formulations is visible on the path $0-1-2-3$. For $F=\{01,23\}$ the two candidates conflict through 12, even though $12\notin F$. Deleting 12 changes the question. Consequently, applying a full-edge-set theorem to the graph with edge set F is not a proof of Theorem 1.1. Likewise, a large induced matching chosen without regard to F may avoid F entirely.

The private-neighbourhood and exchange ideas used here have methodological precedents; see, for example, the local-search analysis of Fürst, Leichter and Rautenbach [3]. The strong edge-colouring bound 21 of Huang, Santana and Yu [2] directly gives the analogous prescribed-set bound with 21. The argument below proves the bound 20 by clique estimates and counting. It does not construct a strong edge-colouring with 20 colours or prove a fractional strong chromatic bound of 20. In general a Hall-ratio bound cannot be promoted to the corresponding fractional chromatic bound [4].

2. Edge covers and two elementary clique bounds

A **conflict clique** is a set $Q\subseteq E(G)$ of pairwise conflicting edges. Throughout the structural part of the proof, Q denotes such a clique and

$$q=|Q|,\qquad d_Q(x)=|\{e\in Q: x\in e\}|.$$

The number $d_Q(x)$ is the degree in the selected edge set, whereas $d_G(x)$ counts all original edges. These quantities will not be interchanged. For $S\subseteq V(G)$, let $e_G(S)$ be the number of original edges with both endpoints in S.

Lemma 2.1 (Endpoint-neighbour cover). For $uv\in E(G)$ put

$$C(uv)=(N_G(u)\setminus\{v\})\cup(N_G(v)\setminus\{u\}).$$

Then $u,v\notin C(uv)$, $|C(uv)|\le 6$, and

$$c(uv)=\sum_{x\in C(uv)} d_G(x)-e_G(C(uv))\le 24-e_G(C(uv)). \tag{2.1}$$

For $uv\in Q$ one also has

$$q\le 1+\sum_{x\in C(uv)} d_Q(x). \tag{2.2}$$

Proof. An edge other than uv conflicts with uv exactly when it has an endpoint in $C(uv)$. Indeed, an edge sharing u or v has its other endpoint in that set; for a disjoint conflicting edge the cross-adjacent endpoint lies in the set. The converse follows immediately from the definition of $C(uv)$. The incidence sum counts an edge meeting $C(uv)$ once, with one extra count precisely when both endpoints lie there. This proves the identity. Each of the two sets forming the cover has at most three vertices. Applying the same cover to $Q\setminus\{uv\}$ proves (2.2). ■

Lemma 2.2 (Triangles and local slack). All edges have $c(e)\le 24$. An edge in a triangle has $c(e)\le 20$. If $c(uv)\ge t$, then $e_G(C(uv))\le 24-t$. If a triangle edge has $c(uv)\ge 20$, its cover has no internal edge. If $c(uv)\ge 21$, then $d_G(u)=d_G(v)=4$ and uv is in no triangle.

Proof. A common neighbour of u,v reduces the cover size to at most five, so (2.1) gives the triangle assertion and the empty-cover assertion at equality. If either endpoint has degree at most three, the cover likewise has size at most five. The other statements follow from (2.1). ■

Lemma 2.3 (Selected support degree at most three). If $d_Q(x) \le 2$ for every x, then $q \le 13$. If $d_Q(x) \le 3$ for every x, then $q \le 18$.

Proof. The empty clique is immediate. Otherwise (2.2) gives $q \le 1 + 6s$ when all selected degrees are at most s. It remains to exclude $q = 19$ for $s = 3$.

Suppose $q = 19$. For every selected edge uv, equality throughout (2.2) forces $|C(uv)| = 6$ and $d_Q(a) = 3$ for every $a \in C(uv)$. We show that every vertex supporting a selected edge has selected degree three. Let $xy \in Q$. Since its cover has size six, there is a neighbour $a \ne y$ of x. This vertex lies in $C(xy)$ and hence has three selected incident edges. Choose a selected edge ab with $b \ne x$. Now $x \in C(ab)$, so $d_Q(x) = 3$ as well. Thus every selected degree is either zero or three. The handshake identity $\sum_x d_Q(x) = 2q = 38$ would make 38 divisible by three, a contradiction. ■

The next lemma will be used inside a set of vertices while retaining every original edge between those vertices.

Lemma 2.4 (A degree-two restriction). Suppose $P \subseteq V(G)$, every edge of a conflict clique Q' has both endpoints in P, and $|N_G(x) \cap P| \le 2$ for every $x \in P$. Then $|Q'| \le 5$. If there is a set $D \subseteq P$ such that each selected edge in Q' has exactly one endpoint in D, then $|Q'| \le 4$. Original edges outside Q' need not respect this bipartition.

Proof. Conflicts between edges with endpoints in P are unchanged in the vertex-induced graph $G[P]$. In that graph an endpoint-neighbour cover has at most two vertices, each of selected degree at most two. Hence $|Q'| \le 1 + 2 \cdot 2 = 5$.

If equality holds, every cover of a selected edge has two vertices of selected degree two. For a supported vertex x choose $xy \in Q'$. Equality forces x to have a neighbour $a \ne y$ in P; this neighbour lies in the cover and has two selected edges. Choose $ab \in Q'$ with $b \ne x$. Since x belongs to the cover of ab, it too has selected degree two. Thus every supported vertex has selected degree two. If each selected edge crosses D, then $|Q'| = \sum_{x \in D} d_{Q'}(x)$ is even, contradicting $|Q'| = 5$. ■

3. Root notation and counting tools

Choose a vertex r, called the root, and write

$$A = N_G(r), \qquad R = \{a \in A : ra \in Q\}, \qquad s = |R| = d_Q(r).$$

When A is independent, every selected edge has exactly one of the following forms: a root edge ra with $a \in R$; an edge ax with $a \in A$ and $x \notin A \cup \{r\}$; or an edge with both endpoints outside $A \cup \{r\}$. Denote the second set by K, the third by O, and put $k = |K|$, $o = |O|$. Thus

$$q = s + k + o, \qquad s + k = \sum_{a \in A} d_Q(a). \tag{3.1}$$

Edges in O will be called outer edges. Even without independence of A, every selected edge either meets A or is outer, so

$$q \le \sum_{a \in A} d_Q(a) + o. \tag{3.2}$$

For $x \ne r$ define its **type**

$$T(x) = N_G(x) \cap A.$$

A vertex of type size one is a singleton, of type size two a double, and of type size three a triple. Let $P = \{x \ne r : |T(x)| \ge 2\}$ be the set of multiple vertices. When A is independent, any vertex of nonempty

type is outside $A \cup \{r\}$. Let B consist of the vertices x incident with an edge $ax \in K$; these are the selected tips.

Lemma 3.1 (Root identities). Assume A is independent. For $xy \in O$,

$$R \subseteq T(x) \cup T(y). \tag{3.3}$$

Moreover,

$$d_O(x) + |T(x)| \le 4, \qquad \sum_{x \in P} |T(x)| \le 3|A| \le 12, \qquad |P| \le 6. \tag{3.4}$$

If all edges of Q have conflict degree at least t, then, for every $a \in R$,

$$L_a := \sum_{x \ne r,\ a \in T(x)} (|T(x)| - 1) \le 24 - t. \tag{3.5}$$

Proof. The outer edge xy conflicts with every selected ra. Its endpoints are not adjacent to r and are distinct from r, a, so at least one is adjacent to a. This proves (3.3).

The neighbours counted by $d_O(x)$ lie outside A, disjoint from those in $T(x)$, proving the first inequality in (3.4). Each label $a \in A$ has at most three neighbours other than r. Counting incidences between A and P proves the other two inequalities.

For (3.5), the sets $A \setminus \{a\}$ and $N_G(a) \setminus \{r\}$ are disjoint, because A is independent. For each $x \ne r$ adjacent to a and each $b \in T(x) \setminus \{a\}$, the original edge bx lies inside $C(ra)$. All these edges are distinct: their endpoints lie in the two disjoint sets just described. Their number is L_a. Lemma 2.2 therefore bounds it by $24 - t$. ■

The summation with two conditions in (3.5) runs over all original vertices of the specified type, not merely selected tips. This is essential when a later outer edge uses a vertex whose edges to A were not selected.

Lemma 3.2 (Independence around a high-degree root). Suppose every selected edge has conflict degree at least 21 and $s \ge 3$. Then $|A| = 4$ and A is independent.

Proof. A selected edge incident with r gives $d_G(r) = 4$ by Lemma 2.2. Every pair of vertices of the four-element set A has at least one endpoint in R, since $|R| \ge 3$. An edge inside A would therefore form a triangle with a selected edge incident with r, contradicting Lemma 2.2. ■

Lemma 3.3 (A singleton controls the selected tips). Assume A is independent. If $ax \in K$ and $T(x) = \{a\}$, then every $y \in B$ with $a \notin T(y)$ is adjacent to x. Consequently $|B| \le 6$.

Proof. Choose $by \in K$. As $a \notin T(y)$, we have $b \ne a$. The edges ax and by have no common endpoint. Independence of A, the singleton type of x, and the exclusion of a from $T(y)$ rule out, respectively, the cross-edges ab, xb and ay. Their required conflict must therefore be witnessed by xy. Thus

$$B \subseteq (N_G(a) \setminus \{r\}) \cup (N_G(x) \setminus \{a\}),$$

whose two sets each have size at most three. ■

Lemma 3.4 (Fully selected doubles). Assume A is independent. Call a double x fully selected if both edges from x to $T(x)$ belong to K, and let D_2 be the set of such vertices. Then $|D_2| \le 4$.

Proof. If two fully selected doubles x, y have different types, choose $a \in T(x) \setminus T(y)$ and $b \in T(y) \setminus T(x)$. Conflict between ax and by forces xy, since all other possible witnesses are excluded by the two type differences and independence of A.

If all vertices of D_2 have the same type, choose a label in that type. Its at most three neighbours other than r contain D_2, so $|D_2| \le 3$. Otherwise choose $x, y \in D_2$ with different types. Every $z \in D_2$ has a

type different from at least one of $T(x), T(y)$, so is adjacent to at least one of x, y. This includes $z = x$ or $z = y$, since xy is an edge. All such vertices lie outside A. Each of x, y has at most two neighbours outside A, yielding $|D_2| \le 4$. ■

Lemma 3.5 (Complementary doubles on outer edges). Suppose $s = |A| = 4$, A is independent, and both endpoints of every edge in $O' \subseteq O$ have type size at most two. Then $|O'| \le 4$.

Proof. Equation (3.3) implies that the endpoint types of each edge in O' are disjoint two-element sets whose union is A. Both endpoints lie in P. Every vertex of P has at least two neighbours in A, so it has at most two neighbours in P, because $A \cap P = \varnothing$. Fix $a \in A$ and put $D = \{x \in P : a \in T(x)\}$. Each edge of O' has exactly one endpoint in D. Lemma 2.4, applied with the full original graph on P, proves the bound. ■

Lemma 3.6 (All types at most two). Suppose $s = |A| = 4$, A is independent, and $|T(x)| \le 2$ for all $x \ne r$. Then $q \le 18$.

Proof. If there is a selected singleton tip, Lemma 3.3 gives $|B| \le 6$. Otherwise $B \subseteq P$, so (3.4) gives the same bound. Every selected tip supports one edge of K, with a second edge possible only for a fully selected double. Therefore $k \le |B| + |D_2| \le 10$ by Lemma 3.4. Lemma 3.5 gives $o \le 4$. Equation (3.1) yields $q \le 4 + 10 + 4 = 18$. ■

4. The five structural bounds

We now prove the estimates used in the packing argument. All degrees in the hypotheses of the following theorem are full degrees in H_G, irrespective of the selected clique.

Theorem 4.1 (Clique bounds by full conflict degree). Let $\Delta(G) \le 4$ and let Q be a conflict clique. For each row of the following table, the condition $c(e) \ge t$ for every $e \in Q$ implies $|Q| \le b_t$.

Threshold t	**20**	**21**	**22**	**23**	**24**
Bound b_t	19	18	17	16	13

The cases are proved in the next five subsections. They use no regularity, girth, or nonemptiness hypothesis on the original graph.

4.1. Conflict degree at least 24

Proposition 4.2. If every $e \in Q$ has $c(e) \ge 24$, then $q \le 13$.

Proof. Equation (2.1) shows that a selected edge has a six-vertex cover with no internal edge. Such an edge belongs to no triangle and no four-cycle: a triangle reduces the cover size, and the edge opposite it in a four-cycle lies inside its cover.

Fix a vertex u and consider selected edges ax with $a \in N_G(u)$ and $x \ne u$. We call all such selected edges with a fixed a a branch at u. Each branch has at most three edges. We claim that a branch containing two edges excludes any nonempty branch with a different label.

Suppose ax_1, ax_2, by are selected, where a, b are distinct neighbours of u and $x_1 \ne x_2$. For either i, the equality $x_i = b$ or $y = a$ would put a selected edge in the triangle uab; equality $x_i = y$ would put ax_i in a four-cycle through u. Thus these selected edges have disjoint endpoints. If ab were an edge, both u and b would belong to $C(ax_i)$, with the original edge ub internal to this cover, which is impossible. The cross-edge ay would form a four-cycle $u - a - y - b - u$ containing by, and bx_i would form a four-cycle containing ax_i. All are excluded. Their conflict must be witnessed by $x_i y$. But then $a - x_1 - y - x_2 - a$ is a four-cycle containing a selected edge, again impossible.

Hence the union of branches with any prescribed set of at most three labels has at most three edges: either all nonempty branches are singletons, or only one branch is nonempty and it has size at most three.

If Q is nonempty, choose $uv \in Q$. Besides uv, there are at most three selected edges incident with u and at most three incident with v. Every remaining selected edge meets $C(uv)$ by Lemma 2.1. Those meeting $N_G(u) \setminus \{v\}$ belong to at most three branches at u and number at most three; the analogous family at v also numbers at most three. Counting these covering families, with any overlaps harmless, gives $q \le 1 + 3 + 3 + 3 + 3 = 13$. ■

4.2. Conflict degree at least 23

Proposition 4.3. If every $e \in Q$ has $c(e) \ge 23$, then $q \le 16$.

Proof. If all selected degrees are at most two, Lemma 2.3 gives $q \le 13$. Otherwise take a vertex r of maximum selected degree, so $s = 3$ or $s = 4$. Lemma 3.2 gives an independent four-element neighbourhood A. Equation (3.5) gives $L_a \le 1$ for each $a \in R$.

No type has size at least three. Indeed, such a type meets R, and at a common label it contributes at least two to the load. Also, two distinct multiple vertices cannot have a common label in R, since they would contribute at least two to its load.

We first show $o \le 2$. If there is an outer edge xy, (3.3) and $|R| \ge 3$ imply that one endpoint, say x, contains at least two labels a, b of R in its type. Consider an outer edge avoiding x. To cover a and b, its endpoints must include vertices having these labels in their types. Any such vertex other than x must be a singleton, by the preceding load observation. Thus the two endpoints would have types $\{a\}$ and $\{b\}$, leaving a third label of R uncovered. This is impossible. All outer edges therefore meet x, and (3.4) bounds their number by two.

If $s = 3$, maximality of the selected degree gives $\sum_{a \in A} d_Q(a) \le 12$. Equation (3.2) then gives $q \le 14$.

If $s = 4$, each label in $A = R$ occurs in the type of at most one multiple vertex. The total number of selected edges from A to multiple vertices is therefore at most four. If singleton tips occur, their number is at most six by Lemma 3.3; otherwise their contribution is zero. Hence $k \le 6 + 4 = 10$, and $q \le 4 + 10 + 2 = 16$. ■

4.3. Conflict degree at least 22

Proposition 4.4. If every $e \in Q$ has $c(e) \ge 22$, then $q \le 17$.

Proof. The case of maximum selected degree at most two follows from Lemma 2.3. Otherwise choose a root of maximum selected degree $s \in \{3, 4\}$. Its neighbourhood is independent and has four vertices. For every selected root label, (3.5) gives $L_a \le 2$.

Case 1: $s = 4$. If some vertex x has $|T(x)| \ge 3$, its type has exactly three elements, since a type of size four exceeds the load bound at each of its labels. Any other multiple vertex has a type meeting $T(x)$; at a common label their load is at least $2 + 1 = 3$, impossible. Thus x is the only multiple vertex. Every outer edge covers four labels and must meet a multiple vertex, hence must meet x. It follows that $o \le 1$. Since at most sixteen selected edges meet A, (3.2) gives $q \le 17$.

We may therefore assume all types have size at most two. Every label occurs in at most two multiple types. The sum of the type sizes is at most eight, giving $|P| \le 4$. Both endpoints of every outer edge belong to P, and each has outer degree at most two. The handshake count gives $2o \le 2|P| \le 8$, hence $o \le 4$.

If there is no selected singleton tip, then $k \le 2|P| \le 8$, giving $q \le 16$. Otherwise choose a selected singleton ax. Lemma 3.3 gives $|B| \le 6$, and each multiple selected tip contributes at most one extra edge beyond its first. Therefore $k \le |B| + |P| \le 10$, so $q \le 18$.

Suppose equality $q = 18$ holds. The preceding bounds force $k = 10$, $o = 4$ and $|P| = 4$. Every vertex of P then has outer degree exactly two. Together with its two neighbours in A, these exhaust all four possible neighbours. In particular no vertex in P is adjacent to the singleton x, since the two outer neighbours already belong to P. Lemma 3.3 therefore implies that every multiple selected tip has a in its type. At most two multiple vertices contain a, by $L_a \le 2$. There can be at most two extra selected edges beyond the first edge at each tip, so $k \le |B| + 2 \le 8$, contradicting $k = 10$. Thus $q \le 17$.

Case 2: $s = 3$. Call a vertex heavy if its type contains at least two labels of R, and write Z for the set of heavy vertices. A heavy vertex contributes at least one unit to L_a at each of at least two labels $a \in R$. Since $\sum_{a \in R} L_a \le 6$, there are at most three heavy vertices. Every outer edge covers the three labels of R, so has a heavy endpoint. Each heavy vertex has outer degree at most two. Hence $o \le 2|Z| \le 6$. The selected degree bound three gives $\sum_{a \in A} d_Q(a) \le 12$, and consequently $q \le 18$.

Suppose again that $q = 18$. Equality forces $o = 6$, $|Z| = 3$, and outer degree two at each heavy vertex. Therefore every heavy vertex has type size exactly two, both labels lying in R. Its contributions to the three loads total two. The three heavy vertices saturate all three load bounds, so every label of R occurs in exactly two heavy types.

Every multiple type meets R, since $|A \setminus R| = 1$. A multiple vertex outside Z would add positive load at a label already saturated by Z. Thus all multiple vertices are heavy. Each heavy vertex has two selected outer edges; because every selected degree is at most three, it supports at most one edge of K. A singleton tip also supports at most one edge of K. If singleton tips occur, Lemma 3.3 now gives $k \le |B| \le 6$. If none occurs, $B \subseteq Z$ gives $k \le 3$. In either event $q = s + k + o \le 3 + 6 + 6 = 15$, contrary to $q = 18$. This excludes equality and proves $q \le 17$. ■

4.4. Conflict degree at least 21

Proposition 4.5. If every $e \in Q$ has $c(e) \ge 21$, then $q \le 18$.

Proof. If every selected degree is at most three, Lemma 2.3 applies. Otherwise choose r with $s = 4$. Its four-element neighbourhood is independent, and (3.5) gives $L_a \le 3$ for all $a \in A$.

If every type has size at most two, use Lemma 3.6. Otherwise choose x with $|T(x)| \ge 3$. There is at most one such vertex, because two types of size at least three intersect, contributing at least four at a shared label. All other multiple types have size two. At a label of $T(x)$, there is at most one multiple vertex other than x: two would contribute at least $2 + 1 + 1 = 4$.

An outer edge incident with x is one of at most one such edges, by (3.4). If there is an outer edge yz avoiding x, its endpoint types are complementary doubles by (3.3). Every multiple vertex $w \ne x$ has a type meeting $T(x)$. A label in that intersection belongs to exactly one of $T(y), T(z)$. The preceding uniqueness at that label forces $w = y$ or $w = z$. Thus y, z are the only possible multiple vertices other than x. Every outer edge avoiding x has two multiple endpoints and consequently must be the single edge yz. There are at most two outer edges in total. At most sixteen selected edges meet A, so $q \le 16 + 2 = 18$. ■

4.5. Conflict degree at least 20

Proposition 4.6. If every $e \in Q$ has $c(e) \geq 20$, then $q \leq 19$.

Proof. If all selected degrees are at most three, Lemma 2.3 gives the stronger bound eighteen. Otherwise choose a root r with $s = 4$, so $A = R$ has four vertices. We may assume $q \geq 20$ and derive a contradiction.

We first prove that A is independent; the threshold twenty alone does not exclude triangles. Suppose ab is an edge inside A. The selected edge ra lies in a triangle, so Lemma 2.2 says that $C(ra)$ has no internal edge. If $x \neq r$ and $a \in T(x)$, any $c \in T(x) \setminus \{a\}$ would give an internal edge cx of $C(ra)$. Hence $T(x) = \{a\}$. Interchanging a, b gives the corresponding assertion for b. An outer edge must cover all four labels of R. To cover a, b, its endpoints would have to have the singleton types $\{a\}$ and $\{b\}$, which cannot cover four labels. Thus $o = 0$, and (3.2) gives $q \leq 16$, a contradiction. This proves independence.

Equation (3.5) now supplies the load bound $L_a \leq 4$ for every $a \in A$. We distinguish all possible sizes of the types.

Case 1: a type has size four. Choose x with $T(x) = A$. Any other type of size at least three would contribute, together with x, at least $3 + 2 = 5$ at a shared label. Thus every other multiple vertex is a double. At each label there is at most one such other multiple vertex, since x already contributes three units.

No outer edge meets x, as all its neighbours lie in A. If an outer edge yz exists, its endpoint types are complementary doubles. Any other multiple vertex has a label in common with one of these types, and uniqueness at that label forces it to be y or z. Therefore yz is the only possible outer edge. Equation (3.2) gives $q \leq 17$, a contradiction.

Case 2: there are two triple types and no type of size four. Let x, y be two distinct triples. Their types share at least two labels; choose distinct shared labels a, b. At either shared label, x and y already contribute $2 + 2 = 4$. Hence any other vertex containing that label in its type is a singleton.

An outer edge avoiding x, y would have to cover a, b using two singleton endpoints of types $\{a\}, \{b\}$. This fails to cover the other two labels. Therefore every outer edge meets x or y. Each has at most one neighbour outside A, so $o \leq 2$. Again $q \leq 16 + 2 = 18$, a contradiction.

Case 3: exactly one triple type and no type of size four. Let x be the unique triple, and let a_0 be the label in $A \setminus T(x)$. Every outer edge avoiding x has complementary double endpoint types. Lemma 3.5 bounds these edges by four. At most one outer edge meets x, giving $o \leq 5$.

We claim $k \leq 10$. The total number of original edges from A to vertices other than r is at most twelve. If at most one of the three edges between x and $T(x)$ is selected, at least two of these original edges are absent from K, proving $k \leq 10$.

Otherwise choose two selected edges ax, bx with distinct $a, b \in T(x)$. Consider any selected edge $a_0y \in K$. We have $y \neq x$, and the type of y has size at most two. As it contains a_0, it misses at least one of a, b, say a. The selected edges ax and a_0y have no common endpoint. Their possible cross-edges aa_0, xa_0 and ay are excluded by independence of A and by the types. Their conflict forces xy. But x has at most one neighbour outside A, so there is at most one possible selected edge $a_0y \in K$. Each of the other three labels supports at most three edges of K. Therefore $k \leq 1 + 3 + 3 + 3 = 10$, as claimed.

Now (3.1) gives $q \leq 4 + 10 + 5 = 19$, contradicting $q \geq 20$.

Case 4: every type has size at most two. Lemma 3.6 gives $q \leq 18$, the final contradiction. These four cases exhaust all type configurations, completing the proof. ■

Propositions 4.2-4.6 prove every row of Theorem 4.1. In particular, the last row gives thirteen, although the final counting argument only needs fifteen.

5. Private neighbourhoods and the packing argument

We temporarily work with an arbitrary finite simple graph J. Closed neighbourhoods are denoted by $N_J[v]$. A maximum independent set means one of largest cardinality, not merely an inclusion-maximal set.

Lemma 5.1 (Optimal independent set and private vertices). Choose a maximum independent set I in J for which $\sum_{v \in I} d_J(v)$ is as small as possible. For $v \in I$ define

$$P_v = \{x \in V(J) : N_J[x] \cap I = \{v\}\}.$$

Then $v \in P_v \subseteq N_J[v]$, the set P_v is a clique, and

$$d_J(x) \geq d_J(v) \quad \text{for every } x \in P_v. \tag{5.1}$$

Proof. A choice with both optimization properties exists because there are finitely many subsets. Since I is independent, v is private to itself. The defining intersection implies $x = v$ or x is adjacent to v, proving the containment.

If distinct $x, y \in P_v$ were nonadjacent, neither could equal v. Replacing v by x, y would give an independent set larger than I, because each has no neighbour in $I \setminus \{v\}$. Thus P_v is a clique.

For $x \in P_v \setminus \{v\}$, the set $(I \setminus \{v\}) \cup \{x\}$ is independent and has the same size as I. Minimality of the degree sum then implies $d_J(v) \leq d_J(x)$. The case $x = v$ is immediate. ■

Lemma 5.2 (Private-neighbourhood count). For a maximum independent set I and the associated private sets,

$$2|V(J)| \leq \sum_{v \in I} (d_J(v) + 1 + |P_v|). \tag{5.2}$$

Proof. Put $t(x) = |N_J[x] \cap I|$. Maximality of I gives $t(x) \geq 1$ for every vertex. The sets P_v are pairwise disjoint and their union consists exactly of the vertices with $t(x) = 1$. A vertex in that union contributes one to $t(x)$ and one to the private-set count; every other vertex contributes at least two to $t(x)$. Thus

$$2|V(J)| \leq \sum_{x \in V(J)} t(x) + \sum_{v \in I} |P_v|.$$

Double-counting pairs (x, v) with $v \in N_J[x] \cap I$ gives $\sum_x t(x) = \sum_{v \in I} (d_J(v) + 1)$, which proves (5.2). ■

Lemma 5.3 (A general twenty-packing criterion). Suppose $\Delta(J) \leq 24$. Assume that every clique whose vertices all have degree at least t has size at most b_t, where

$$b_{20} = 19, \quad b_{21} = 18, \quad b_{22} = 17, \quad b_{23} = 16, \quad b_{24} = 15.$$

Then $\alpha(J) \geq |V(J)|/20$.

Proof. Take I as in Lemma 5.1. For $v \in I$ write $d = d_J(v)$. If $d \leq 19$, the containment $P_v \subseteq N_J[v]$ gives

$$d + 1 + |P_v| \leq 2(d + 1) \leq 40.$$

If $20 \leq d \leq 24$, the private set is a clique all of whose vertices have degree at least d, by (5.1). Hence $|P_v| \leq b_d = 39 - d$, and again $d + 1 + |P_v| \leq 40$. Substituting into (5.2) yields $2|V(J)| \leq 40|I|$. Since $|I| = \alpha(J)$, this is the assertion. ■

Proof of Theorem 1.1. Form H_G using the full original graph and let $J = H_G[F]$. Lemma 2.1 gives $\Delta(H_G) \le 24$, and therefore $\Delta(J) \le 24$.

Let Q be a clique in J whose vertices have degree at least t in J, where $20 \le t \le 24$. It is also a clique in H_G, and for each $e \in Q$ we have $c(e) = d_{H_G}(e) \ge d_J(e) \ge t$. Theorem 4.1 applies. Its five bounds meet those in Lemma 5.3, with the stronger last bound $13 \le 15$. Thus J has an independent set M with $|F| = |V(J)| \le 20|M|$. Its vertices are edges from F, and their independence in J says that they are pairwise nonconflicting in the original graph. Therefore M is the required induced matching in G. If F is empty, the same argument or the choice $M = \varnothing$ applies. Sharpness is proved next. ■

6. Sharpness and the precise scope of the conclusion

Proposition 6.1 (Sharpness). There is a 4-regular graph B with twenty edges for which every induced matching has size at most one.

Proof. Start with a cycle of length five. Replace each vertex by an independent pair V_i, indexed modulo five, and join every vertex of V_i to every vertex of V_{i+1}. There are ten vertices, each with four neighbours, and twenty edges.

Every edge belongs to one of the five complete bipartite graphs between consecutive pairs. Consider two distinct edges. If they belong to the same such graph, they either meet or have cross-adjacent endpoints. If their corresponding cycle edges meet, an endpoint in their common cluster is adjacent to an endpoint of the other edge in a neighbouring cluster. If the corresponding cycle edges are disjoint, some endpoint cluster of one is adjacent on the five-cycle to an endpoint cluster of the other, again giving cross-adjacent endpoints. Thus every pair conflicts. A single edge is an induced matching, so $\nu_s(B) = 1$. Taking $F = E(B)$ in an inequality $|F| \le C|M|$ forces $C \ge 20$. ■

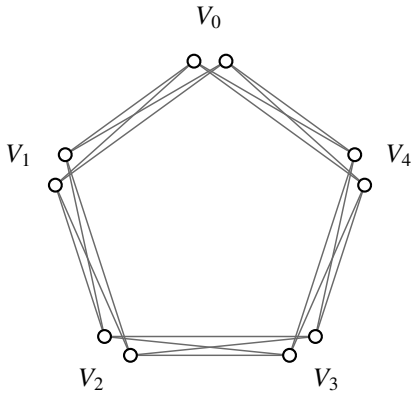


Figure 1. The doubled five-cycle: five independent pairs, ten vertices and twenty edges. Each pair of consecutive clusters induces a complete bipartite graph.

This is the doubled five-cycle already used by Joos and Nguyen [1]; neither the example nor the sharpness of the full-edge-set constant is new here.

The proof establishes a cardinality guarantee separately for each candidate set. It does not establish the analogous inequality for every assignment of arbitrary nonnegative edge weights. In conflict-graph language, the conclusion is the Hall-ratio bound (1.2). The general inequalities $\rho(J) \le \chi_f(J) \le \chi(J)$ point in the opposite direction from a proposed deduction of a fractional or ordinary colouring bound from a Hall-ratio bound. Known separations between Hall ratio and fractional chromatic number [4] reinforce the need to keep these statements distinct.

The optimization used in Lemma 5.1 is an existence argument. No running-time claim about the accompanying Lean theorem is made. The mathematical proof is independent of a particular implementation of maximum-independent-set search.

Appendix A. Formalization of this manuscript

This manuscript retains the mathematical proof of the earlier paper-first exposition. Its corresponding entry point is now **PaperRoute.induced_matching_packing20**, rather than the original **StrongEdge** endpoint. The two developments share mathematical ingredients, but this route explicitly proves the stronger intermediate statements used in the exposition. This is a correspondence between mathematical arguments and formal declarations, not a claim that the prose itself is machine-checked.

The file **ExactCover.lean** proves both directions of the cover characterization and the exact additive identity $c(e) + e_G(C(e)) = \sum_{x \in C(e)} d_G(x)$. Natural subtraction in (2.1) is derived from that identity. **UniformLoad.lean** proves one theorem with an arbitrary natural threshold t. Its more general internal interface assumes disjointness of the two erased neighbour sets; independence of A supplies that hypothesis in (3.5). The specialized root-load interfaces used by the supporting local proofs instantiate this uniform argument.

Foundations.lean proves both exact identities in (3.1), the unspecialized incidence estimate in (3.4), and the triangle degree bound. **Thresholds.lean** assembles Propositions 4.2-4.6, uses the same threshold-free small-type lemma in the 21 and 20 cases, and establishes $o \le 1$ and $q \le 17$ in the four-element-type case of Proposition 4.6. **Packing.lean** contains the five-field criterion, its induced-subgraph transfer and the final theorem. **PrivateSemantics.lean** identifies the code's private-vertex predicate with the singleton closed-neighbourhood intersection in Section 5. **Sharpness.lean** proves the doubled-cycle witness and the lower bound on an arbitrary real constant.

Manuscript statement	Declaration or module in PaperRoute
Exact cover, (2.1)	ExactCover: conflict_iff_meets_cover; conflict_degree_add_inside_eq_sum
Uniform slack and load, (3.5)	inside_cover_budget; UniformLoad: root_load_budget
Exact root partition, (3.1)	Foundations: selected_card_eq_star_root_outer; root_incidence_eq_star_add_root
Lemmas 2.3-2.4	Library/Bounds/SubcubicEighteen; DegreeTwoClique
Lemmas 3.3-3.4	Library/Bounds/RootCounting; High21Double
Lemma 3.5, arbitrary outer subset	OuterSubsets: outer_subset_card_le_four
Lemma 3.6	Thresholds: small_type_clique_bound
Propositions 4.2-4.6	Thresholds: proposition_4_2 through proposition_4_6
Stronger four-type case	Thresholds: outer_card_le_one_of_type_four; clique_card_le_seventeen_of_type_four
Lemmas 5.1-5.3	Library/Basic; PrivateExchange; DoubleCount; Packing; PaperRoute/Packing
Theorem 1.1	Packing: induced_matching_packing20
Proposition 6.1, real sharpness	Sharpness: doubleC5_matching_card_le_one; twenty_is_necessary_real

The directory **PaperRoute/Library** began as an explicitly recorded namespace fork of the original elementary and local proofs. Its supporting files were then refactored where this route needs exact counting and the unified budget. These shared arguments are not presented as independently discovered proofs. The manuscript-level propositions and the final packing theorem have their own proof terms. The dependency audit checks that this endpoint does not call any **StrongEdge** declaration or the copied library's final packing theorem. The two root-count identities also have their own proofs, even where a later argument only needs their inequality consequences.

The project uses Lean **4.35.0-rc4** and mathlib revision **49fd9dee36e7bbf345fe7981d2de6aeeb591f75b**. The delivery includes build logs, exact source hashes, a declaration-dependency audit and a numbered crosswalk. The accepted foundational axioms are **propext**, **Classical.choice** and **Quot.sound**. The companion Route B manuscript follows the unchanged original formal development. Neither route claims a different global strategy or a new historical priority.

References

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[2] M. Huang, M. Santana and G. Yu, *Strong Chromatic Index of Graphs With Maximum Degree Four*, Electronic Journal of Combinatorics 25 (2018), no. 3, Paper P3.31. DOI: 10.37236/7016.

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